

Rodrigo Fernandez

Senior Java (Spring Boot) Backend Engineer

Katy, Texas | +1 (407) 801-1287 | <https://www.linkedin.com/in/rodrigo-bermejo-fern%C3%A1ndez-9a5a0664/>
rodrigobfdev@gmail.com

Summary

Senior Java Backend Engineer with strong experience across fintech, cloud infrastructure, SaaS, e-commerce, and enterprise workflow platforms, using **Java 8–17**, **Spring Boot 2.x–3.x**, **Spring Security 5.x–6.x**, **Hibernate 5.x–6.x**, **PostgreSQL 10–15**, **Kafka 2.x–3.x**, **AWS**, and Docker-based delivery. Specialized in designing reliable REST APIs, event-driven services, secure authentication flows, database optimization, legacy migration, and production debugging, with practical ownership of high-volume systems, performance tuning, CI/CD stabilization, observability, and cross-team backend delivery for regulated, customer-facing products worldwide.

Skills

- **Backend:** Java 8–17, Spring Boot 2.x–3.x, Spring MVC 4.x–6.x, Spring Security 5.x–6.x, Spring Data JPA, Hibernate 5.x–6.x, REST APIs, Microservices, Batch Processing, OAuth2, JWT, RBAC
- **Database:** PostgreSQL 10–15, MySQL 5.7–8.x, Oracle Database 11g–19c, Redis 5–7, SQL optimization, indexing, query tuning, transaction isolation, schema migration
- **Messaging & Integration:** Kafka 2.x–3.x, RabbitMQ 3.x, AWS SQS, AWS SNS, Webhooks, Event-driven architecture, retry handling, idempotency, distributed transactions
- **Cloud & DevOps:** AWS EC2, ECS, Lambda, S3, RDS, CloudWatch, Docker 19–24, Kubernetes 1.20–1.28, Jenkins 2.x, GitHub Actions, GitLab CI, Maven, Gradle
- **Testing & Quality:** JUnit 4–5, Mockito 3–5, Testcontainers, Postman, Swagger/OpenAPI 3.x, SonarQube, integration testing, contract testing, load testing
- **Architecture:** Microservices, monolith-to-service migration, DDD, Clean Architecture, API Gateway, caching strategy, observability, CI/CD hardening, production incident resolution

Experience

Lightedge — Senior Software Engineer (Apr 2020 – Mar 2026)

- Architected **Java 11–17** and **Spring Boot 2.7–3.2** backend services for cloud infrastructure and enterprise hosting workflows, improving API reliability by **34%** through stronger validation, retry control, and centralized exception handling.
- Migrated legacy **Spring Boot 2.3** services to **Spring Boot 3.x** with **Jakarta EE**, updated deprecated security filters, and resolved dependency conflicts that were blocking production builds after framework upgrades.
- Designed secure REST APIs with **Spring Security 5.x–6.x**, OAuth2, JWT, and role-based access rules, reducing unauthorized access defects by **28%** across internal customer and operations portals.
- Improved **PostgreSQL 12–15** query performance by rewriting slow joins, adding composite indexes, and using query plans to reduce report generation time from minutes to seconds.
- Built event-driven provisioning workflows with **Kafka 2.x–3.x** and AWS messaging services, allowing infrastructure actions to run asynchronously without blocking customer-facing API responses.
- Resolved a recurring production issue where duplicate messages created inconsistent provisioning states by adding idempotency keys, database constraints, and safe retry handling.

- Implemented health checks, structured logs, and **CloudWatch** dashboards for Java services, helping support teams identify failed integrations faster and reduce incident triage time by **31%**.
- Refactored tightly coupled service classes into cleaner domain and application layers, making the backend easier to test, extend, and maintain across multiple product teams.
- Created automated API regression suites with **JUnit 5**, Mockito, and Testcontainers, increasing backend test coverage and preventing repeated failures during release pipelines.
- Optimized Docker-based deployments by reducing image size, cleaning unused dependencies, and aligning JVM memory settings with container limits for more predictable production behavior.
- Collaborated with frontend, DevOps, and product teams to implement customer account, billing, and infrastructure management features while keeping backend contracts stable.
- Improved CI/CD reliability with Jenkins and GitHub Actions by fixing flaky integration tests, adding environment checks, and standardizing Maven build profiles across services.

NIX United — Software Engineer *(Oct 2016 – Mar 2020)*

- Developed enterprise backend applications with **Java 8–11**, **Spring Boot 1.5–2.3**, and **Hibernate 5.x**, supporting SaaS, finance, logistics, and internal workflow systems.
- Built RESTful APIs for customer onboarding, transaction management, and reporting modules, improving backend response consistency and reducing API-related defects by **25%**.
- Migrated older **Spring MVC 4.x** modules into **Spring Boot 2.x** services, replacing XML-heavy configuration with annotation-based setup and cleaner dependency management.
- Fixed a critical memory leak caused by unclosed database sessions and oversized DTO mapping, improving long-running service stability under peak traffic.
- Designed transactional workflows with **PostgreSQL 10–12** and MySQL 5.7–8.x, using optimistic locking and careful rollback handling to prevent inconsistent business records.
- Implemented asynchronous job processing with RabbitMQ and scheduled Java workers, reducing manual operations work and improving background task completion visibility.
- Improved authentication flows with **Spring Security 5.x**, JWT, password hashing, and permission checks for multi-role users across admin and customer systems.
- Reduced deployment issues by improving Maven build profiles, Docker environment variables, and service startup validation for staging and production releases.
- Added Swagger/OpenAPI 3.x documentation and request examples, helping frontend and QA teams integrate faster with backend endpoints and reduce clarification cycles.
- Debugged API timeout issues by analyzing logs, SQL execution plans, and thread behavior, then optimized database access patterns and service-level caching.
- Worked flexibly across backend development, API design, production support, and database optimization, keeping delivery stable even when project requirements changed quickly.

Webiz Digital — Software Developer *(Jan 2014 – Sep 2016)*

- Built Java backend features for e-commerce, content management, and customer service platforms using **Java 7–8**, **Spring MVC 4.x**, Hibernate, JSP, and REST APIs.
- Implemented order, payment callback, inventory, and notification modules with clear service-layer logic, improving transaction tracking accuracy by **22%** for business users.
- Migrated selected legacy Servlet and JSP flows into **Spring MVC 4.x** controllers, reducing duplicated request handling and making the codebase easier to maintain.
- Fixed a production bug where payment callbacks were processed twice by adding database-level uniqueness, safer transaction boundaries, and backend validation checks.
- Improved MySQL 5.6–5.7 performance by adding indexes, reducing N+1 Hibernate queries, and replacing inefficient dynamic SQL with prepared repository methods.

- Integrated third-party shipping, email, and payment APIs using resilient retry logic, timeout controls, and detailed error logging for faster troubleshooting.
- Created admin backend endpoints for product management, customer records, and reporting dashboards while coordinating API contracts with UI developers.
- Added unit and integration tests with **JUnit 4** and Mockito, helping the team catch regression issues before releases and reduce QA rework.
- Supported Linux-based deployments by reviewing logs, fixing configuration issues, and coordinating application restarts during controlled release windows.
- Used Java backend skills flexibly across bug fixing, new feature delivery, database updates, and integration support for multiple client-facing business systems.

Aurio — Junior Software Developer *(Oct 2012 – Dec 2013)*

- Developed backend modules with **Java 6–7**, Servlets, JSP, Spring 3.x, JDBC, and Hibernate 3.x for internal business and customer management applications.
- Created basic CRUD services, validation logic, and database access layers, helping senior engineers deliver stable features for operational users.
- Fixed form submission and session timeout issues by reviewing server logs, debugging request lifecycle behavior, and improving validation messages.
- Improved SQL queries in MySQL 5.x by removing unnecessary joins and adding clearer filtering logic for faster search and reporting screens.
- Assisted in migrating small legacy JSP pages toward cleaner MVC patterns, separating controller logic from view templates and reusable service methods.
- Added backend support for email notifications, user status updates, and simple approval workflows using Java utilities and scheduled tasks.
- Worked with QA feedback to reproduce bugs, identify root causes, and deliver fixes without breaking existing user flows or database records.
- Maintained deployment notes and configuration details for Linux application servers, reducing repeated setup mistakes during environment refreshes.
- Built a strong foundation in Java backend development by supporting production fixes, feature enhancements, database scripts, and team code reviews.

Microsoft — Software Developer Intern *(Jun 2012 – Sep 2012)*

- Completed a Software Developer Internship focused on Java fundamentals, internal tooling, unit testing, and code review discipline inside a structured enterprise engineering environment.
- Assisted with small bug fixes and documentation tasks, learning how production teams track defects, validate edge cases, and communicate changes across reviewers.
- Built early professional habits around code review, debugging, documentation, testing, and secure engineering practices while contributing to production-adjacent enterprise tools.
- Learned large-scale development workflow, version control discipline, and professional communication from senior engineers.
- Participated in team meetings, sprint planning, and defect analysis while gaining real-world software lifecycle exposure.

Education

Texas State University

Bachelor's Degree in Computer Science *(Sep 2008 – May 2012)*